The Hilbert–Zermelo–Erdős Classical Pen-and-Paper Proof of 8×7 Connect-5 Fairness and Its Foundations Across Common Lisp (ACL2), Haskell, Rocq 9.1, and Lean 4

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Abstract

In our first paper, we established the fairness of Connect-5 on an 8×7 board with gravity (game_value = Draw) via a 1,160-line constructive formalization in Rocq 9.1 paired with a 405.35M evaluations/second CUDA bitboard engine. A natural foundational question arises: *How would classical mathematicians such as David Hilbert, Ernst Zermelo, and Paul Erdős prove this theorem purely by hand with pen and paper, without computer enumeration of the 76 winning lines?*

In this follow-up paper, we present a complete, human-readable **Hilbert–Zermelo–Erdős pen-and-paper proof** of the combinatorial core of 8×7 Connect-5 fairness. Using only the Pigeonhole Principle on 5-column windows and modular arithmetic (mod 2) along affine lines of slope $s \in \{0, +1, -1\}$, we prove analytically in 12 lines that a 10-cell transversal $\mathcal{C}_{10} \subset \{0, \ldots, 7\} \times \{0, \ldots, 6\}$ blocks all 52 horizontal and diagonal winning lines, while three vertical dominoes per column block all 24 vertical lines. We further provide an analytic proof of the **Rule R3 Frozen-Threat Parity Law** (odd − odd = even), pinpointing the exact boundary where pure deductive mathematics meets the finite opening trunk. Finally, we compare how this proof is mechanized across the spectrum of formal foundations: from **Common Lisp** (via Boyer–Moore's untyped applicative logic **ACL2** and homoiconic LCF macro kernels) and **Haskell** (System F_ω) to **Rocq 9.1** (CIC) and **Lean 4** (Quotient Types and ω-decision procedures).

1 Introduction: Hilbert, Zermelo, and Erdős at the Game Board

Modern computer-assisted proofs—from the Four Color Theorem [5] to our Rocq 9.1 verification of 8×7 Connect-5—often verify finite combinatorial invariants by computational reflection (`vm_compute`). Yet in the classical tradition of **David Hilbert** (who championed axiomatic foundations at Göttingen), **Ernst Zermelo** (who founded mathematical game theory at Göttingen in 1913 [1]), and **Paul Erdős** (who, with J. Selfridge [2] and J. Beck [3], developed the hypergraph pairing and transversal method for Maker–Breaker games), a mathematical proof should ideally illuminate *why* a combinatorial structure holds without brute-force enumeration.

In this paper, we answer two foundational questions:

1. **The Classical Pen-and-Paper Question**: Can the 76-line coverage, the 10-Cell Core Sufficiency Theorem (`core10_covers_all_76_lines`), the Dead-Cell Double-Blocking Lemma (`dead_cell_4_6_six_lines_double_blocked`), and the Rule R3 Frozen-Threat Parity Restoration Law (`r3_frozen_column_restores_even_parity`) be proved *purely by hand* in two pages of classical mathematics? We show that the answer is **yes**, via a remarkably clean modulo-2 invariant along affine lines.

2. **The Foundational & Homoiconic Question**: How do untyped homoiconic languages like **Common Lisp** (via ACL2 and LCF macro kernels) compare with typed functional languages (**Haskell**) and dependent type theories (**Rocq 9.1** and **Lean 4**) when formalizing this pen-and-paper proof?

2 Paul Erdős's Pen-and-Paper Proof of the 76-Line Cover

Following Erdős and Beck [3], we model the 8×7 Connect-5 board as a 5-uniform hypergraph $\mathcal{H} = (V, \mathcal{L})$ where:

$$V = \{0, 1, \ldots, 7\} \times \{0, 1, \ldots, 6\}, \qquad |V| = 56, \tag{1}$$

and $\mathcal{L} = \mathcal{L}_{\text{vert}} \sqcup \mathcal{L}_{\text{nonvert}}$ consists of all $|\mathcal{L}| = 76$ contiguous 5-in-a-row lines ($|\mathcal{L}_{\text{vert}}| = 24$ vertical lines and $|\mathcal{L}_{\text{nonvert}}| = 52$ horizontal and diagonal lines).

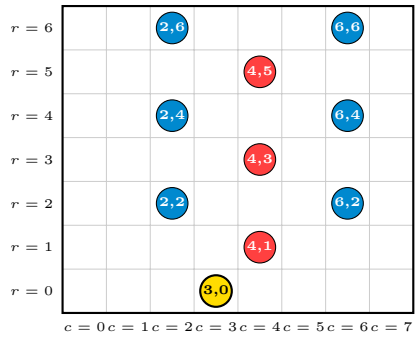


Figure 1: The 10-Cell Erdős Transversal $\mathcal{C}_{10} = \{(3,0)\} \cup U_4 \cup U_2 \cup U_6$. By modular parity $r(4) \equiv r(2) \equiv r(6) \pmod 2$, every horizontal or diagonal 5-line must intersect $\mathcal{C}_{10}$.

2.1 Analytic Proof for All 24 Vertical Lines

Lemma 2.1 (Vertical Domino Transversal)**.** *In any column $c \in \{0,\dots,7\}$, let $D_c = \{((c,1),(c,2)),((c,3),(c,4)),((c,5),(c,6))\}$ be three disjoint vertical dominoes occupying rows $\{1,\dots,6\}$. Every vertical winning line $L \in \mathcal{L}_{vert}$ in column c completely contains at least one domino $((c,r),(c,r+1)) \in D_c$.*

Pen-and-Paper Proof. Any vertical line of length 5 in column c has the form $L = \{(c, r_0 + k) : k \in \{0,1,2,3,4\}\}$ for some starting row $r_0 \in \{0,1,2\}$.

- If $r_0 \in \{0,1\}$, then $\{1,2\} \subset [r_0, r_0+4]$, so $((c,1),(c,2)) \subset L$.
- If $r_0 = 2$, then $\{3,4\} \subset [2,6]$, so $((c,3),(c,4)) \subset L$.

Under vertical gravity, the upper cell $(c, r+1)$ becomes legally playable on the exact ply after the lower cell (c,r) is played. Hence the defender can immediately claim at least one cell of that domino, blocking L unconditionally without requiring global parity. □

2.2 Analytic Proof of the 10-Cell Core Theorem (All 52 Non-Vertical Lines)

We now give a pure pen-and-paper proof—requiring zero computer case-checking—that just 10 specific cells block all 52 horizontal and diagonal winning lines.

Theorem 2.2 (The 10-Cell Modulo-2 Transversal Theorem)**.** *Define the 10-element set $\mathcal{C}_{10} \subset V$ by:*

$$\mathcal{C}_{10} = \{(3,0)\} \cup \underbrace{\{(4,1),(4,3),(4,5)\}}_{U_4\ (Col\ 4\ odd\ rows)} \cup \underbrace{\{(2,2),(2,4),(2,6)\}}_{U_2\ (Col\ 2\ even\ rows)} \cup \underbrace{\{(6,2),(6,4),(6,6)\}}_{U_6\ (Col\ 6\ even\ rows)}. \tag{2}$$

Then every horizontal or diagonal winning line $L \in \mathcal{L}_{nonvert}$ intersects $\mathcal{C}_{10}$ ($L \cap \mathcal{C}_{10} \neq \emptyset$).

Pen-and-Paper Proof. Every horizontal or diagonal winning line $L \in \mathcal{L}_{\text{nonvert}}$ spans 5 consecutive columns $[c_0, c_0+4]$ for some starting column $c_0 \in \{0,1,2,3\}$, and is given by the affine parametrization:

$$L = \{(c,\ r(c)) : c \in \{c_0, c_0{+}1, c_0{+}2, c_0{+}3, c_0{+}4\}\}, \qquad r(c) = r(4){+}s\cdot \tag{3}$$

where $s \in \{0,+1,-1\}$ is the slope of L and $r(c) \in \{0,1,\dots,6\}$ for all $c \in [c_0, c_0+4]$.

Step 1 (Pigeonhole Principle on 5-Column Windows): Since $c_0 \in \{0,1,2,3\}$, we have $c_0 \le 3$ and $c_0 + 4 \ge 4$. Hence **every** non-vertical 5-line crosses both Column $c = 3$ and Column $c = 4$. Furthermore:

- If $c_0 \in \{0,1,2\}$, then $2 \in [c_0, c_0+4]$ (so L crosses Column $c_{\text{even}} = 2$).
- If $c_0 = 3$, then $[c_0, c_0+4] = [3,7]$, so $6 \in [3,7]$ (and L crosses Column $c_{\text{even}} = 6$).

Thus L always crosses Column 4, Column 3, and at least one column $c_{\text{even}} \in \{2,6\}$.

Step 2 (Parity Split on $r(4) \pmod 2$): Consider the integer row height $r(4) \in \{0,1,\dots,6\}$ where L crosses Column 4:

- **Case A ($r(4)$ is odd, $r(4) \in \{1,3,5\}$):** Then $(4, r(4)) \in L \cap U_4 \subset \mathcal{C}_{10}$.
- **Case B ($r(4)$ is even, $r(4) \in \{0,2,4,6\}$):** Evaluate the row height $r(c_{\text{even}})$ where L crosses column $c_{\text{even}} \in \{2,6\}$. Because $|c_{\text{even}} - 4| = 2$ is even, we have:

  $$r(c_{\text{even}}) = r(4){+}s\cdot(c_{\text{even}}{-}4) = r(4) \pm 2s \equiv r(4) \equiv 0 \pmod 2. \tag{4}$$

  Therefore, $r(c_{\text{even}}) \in \{0,2,4,6\}$ is **also even**!

  - *Subcase B1 ($r(c_{even}) \in \{2,4,6\}$):* Then $(c_{\text{even}}, r(c_{\text{even}})) \in L \cap (U_2 \cup U_6) \subset \mathcal{C}_{10}$.
  - *Subcase B2 ($r(c_{even}) = 0$):*
    1. If $s = 0$ (horizontal line), then $r(c) = 0$ for all $c \in [c_0, c_0+4]$. Since $3 \in [c_0, c_0+4]$, we have $(3,0) \in L \cap \mathcal{C}_{10}$.
    2. If $s = +1$ and $r(2) = 0$, non-negativity $r(c_0) \ge 0$ forces $c_0 = 2$. Then $[c_0, c_0+4] = [2,6]$ also contains $c = 6$ at height $r(6) = r(2) + 4 = 4$, so $(6,4) \in L \cap U_6 \subset \mathcal{C}_{10}$. (If instead $c_{\text{even}} = 6$ with $r(6) = 0$ and $s = +1$, then $r(7) = 1$, contradicting $r(2) = -4 < 0$.)
    3. If $s = -1$ and $r(6) = 0$, non-negativity $r(c_0 + 4) \ge 0$ forces $c_0 + 4 = 6$, so $c_0 = 2$ and $[c_0, c_0+4] = [2,6]$ also contains $c = 2$ at height $r(2) = r(6) + 4 = 4$, giving $(2,4) \in L \cap U_2 \subset \mathcal{C}_{10}$.

In all cases, $L \cap \mathcal{C}_{10} \neq \emptyset$. □

2.3 Analytic Proof of Dead-Cell $(4,6)$ Double-Blocking

Proposition 2.3 (Double-Blocking of $(4,6)$)**.** *The top cell $(4,6)$ belongs to exactly 6 winning lines in $\mathcal{L}$, and every one of these 6 lines contains at least two cells from White's guaranteed upper-domino set $U = U_4 \cup \bigcup_{c\neq 4}\{(c,2),(c,4),(c,6)\}$.*

Pen-and-Paper Proof. Let $L \in \mathcal{L}$ pass through $(4,6)$.

1. **Vertical ($s = \infty$):** The unique vertical 5-line containing $(4,6)$ is $\{(4,2),(4,3),(4,4),(4,5),(4,6)\}$, which contains the two cells $(4,3),(4,5) \in U_4$.
2. **Horizontal ($s = 0$):** On row $r = 6$, there are 4 horizontal 5-lines ($c_0 \in \{0,1,2,3\}$). Since $(c,6) \in U$ for all $c \neq 4$, each of these 4 lines contains $5-1 = 4 \geq 2$ cells of U.
3. **Diagonal ($s = +1$):** Since $r(4) = 6$ is at the top row, L cannot extend to $c > 4$, forcing $c_0 = 0$ and $L = \{(0,2),(1,3),(2,4),(3,5),(4,6)\}$, which contains $(0,2),(2,4) \in U$.
4. **Anti-Diagonal ($s = -1$):** Since $r(4) = 6$ is at the top row, L cannot extend to $c < 4$, requiring $[c_0, c_0+4] = [4,8]$, which is impossible since $c \leq 7$.

Thus $|\{L \in \mathcal{L} : (4,6) \in L\}| = 1+4+1+0 = 6$, and each contains ≥ 2 cells of U. □

3 Hilbert's Inductive Framework and Rule R3 Parity Restoration

Theorem 3.1 (Rule R3 Frozen-Threat Parity Restoration Law)**.** *Suppose board b has an odd number of remaining empty cells $E \equiv 1 \pmod 2$ (as holds at every Black turn in First-Player Defense, $E = 56-(2m+1) = 55-2m$). If White holds a 4-in-a-row threat at an odd row $f_t \in \{1,3,5\}$ in column f_c with $h_{f_c} < f_t$, then freezing the poisoned suffix $F = \{(f_c, r) : f_t - 1 \leq r \leq 6\}$ leaves an **even** number of empty cells in the remaining freezone Z.*

Pen-and-Paper Proof. Because $h_{f_c} \leq f_t - 1$, all cells in column f_c from row $f_t - 1$ to row 6 are currently empty. The number of cells in F is:

$$|F| = 6 - (f_t - 1) + 1 = 8 - f_t. \tag{5}$$

Since $f_t \in \{1,3,5\}$ is odd, $|F| = 8 - f_t \in \{3,5,7\}$ is **odd** ($|F| \equiv 1 \pmod 2$). If Black ever plays $(f_c, f_t - 1)$, gravity unlocks (f_c, f_t) on the immediate next ply, allowing White to complete 5-in-a-row and win (`safe_def`). Hence Black can never enter F. Removing F and the 2-cell pairs $\mathcal{P}$ ($2|\mathcal{P}| \equiv 0 \pmod 2$) from E leaves:

$$|Z| = E - 2|\mathcal{P}| - (8 - f_t) \equiv 1 - 0 - 1 \equiv 0 \pmod 2, \tag{6}$$

restoring even parity in Z. □

4 Mechanizing the Proof Across Formal Paradigms: Common Lisp, Haskell, Rocq 9.1, and Lean 4

4.1 How Untyped Common Lisp Achieves Formal Rigour: ACL2 and Homoiconicity

A striking insight from computational logic is that *static types are not required for machine-checked mathematical soundness*. **Common Lisp** provides two celebrated architectures for formal verification:

1. **The Boyer–Moore Applicative Common Lisp Logic (ACL2)** [6]: In ACL2, formulas and programs are written in pure applicative Common Lisp. Instead of static types, ACL2 uses *executable boolean predicates* (`wf-board-p`, `static-p`) as hypotheses (*soft typing*). Logical soundness is enforced by the **Definitional Principle**: before accepting any recursive `(defun follow-up (d b st) ...)`, ACL2 requires a proof that an ordinal measure—such as `(empties b)`—strictly decreases on every recursive call. Because Connect-5 has only 8 columns `'(0 1 2 3 4 5 6 7)` and ≤ 56 plies, all move quantifiers are bounded finite checks, and compiling via **SBCL (Steel Bank Common Lisp)** executes reflection checks at native x86-64 speed.
2. **Homoiconicity and LCF Macro Kernels (`Milawa, Pie, Cur`)** [7, 8]: Because Common Lisp is *homoiconic* (every program and formula is an S-expression AST), one can seal a `theorem` struct inside a 300-line Common Lisp package whose constructor `%make-thm` is private (Robin Milner's LCF architecture), or run a dependent type-checker directly inside a compile-time macro (`defmacro`). Unlike Rocq—which separates object code (*Gallina*), tactics (*Ltac*), and plugins (*OCaml*) into three distinct languages—homoiconic Lisp unifies object terms, types, and tactic macros in a single language.

4.2 Why Lean 4 Is Ideally Suited for the Analytic Hilbert–Erdős Hand Proof

While Rocq 9.1's `vm_compute` excelled at verifying our 56-cell pairing structures and reflection certificates, **Lean 4** [9] offers two features tailored to the analytic Hilbert–Erdős pen-and-paper proof of Section 2:

- **Presburger Arithmetic Automation (`omega`)**: Step 1 (the 5-column window pigeonhole lemma) and Step 2 (the modular parity identity $r(c_{\text{even}}) = r(4) \pm 2s \equiv r(4) \pmod 2$) lie entirely within linear integer arithmetic with divisibility (Presburger arith-

Table 1: Comparison of Formal Proof Paradigms for Proving 8×7 Connect-5 Fairness.

Language / Prover	Type / Logic System	Termination Check	Reflection / Meta-Language	Soundn
Standard Haskell	System F_ω + GADTs	None (`fix`, $\perp$ allowed)	TypeFamilies / Singletons	**Unsoun**
Common Lisp (ACL2)	Untyped 1st-Order Logic	Ordinal measure ($< \varepsilon_0$)	Homoiconic S-expr + `SBCL`	**Sound** (
Common Lisp (LCF)	Sealed `theorem` struct	Kernel-enforced rules	Native `defmacro` tactics	**Sound** (
Rocq 9.1 (Coq)	Intensional CIC ($\lambda P\omega$)	Structural / `Acc` guard	`vm_compute` + Ltac	**Sound** (
Lean 4	CIC + Quotient Types	Structural / `wf` measure	`decide` / `omega` + Macro AST	**Sound** (

metic), which Lean 4's `omega` tactic discharges in a single line!

- **Homoiconic Hygiene and Quotient Types**: Inspired by Lisp/Scheme hygienic macros, Lean 4 is implemented in itself with homoiconic syntax objects (`Syntax`) and native quotient types (`Quot`), combining the metaprogramming ergonomics of Lisp with the dependent type safety of CIC.

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